



Friedrich-Alexander-Universität
Erlangen-Nürnberg

Neutron Activation Analysis

[ALC-B29]

Department of Physics

Lab Report

M766

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Experiment Date: 19.06.2024

Abstract

Neutron Activation analysis is a nuclear analytical technique primarily used to determine a sample's elemental composition. It is done by irradiating the sample material with neutrons, causing it to become radioactive. Resulting in isotope decay and emitting gamma rays, which are then measured and analysed to quantify the elemental composition.

To execute the experiments, we utilise a lithium-doped Germanium semiconductor detector and collect the signal through a Multi-Channel Analyser. This signal is extracted on a readout computer as a plot of the Gamma Spectrum.

On the principle basis of the experiment, we analyse the graph of two samples; Copper and Vanadium. Along with the samples we also analyse the percentage of copper present in a stack of 1 euro coin and the percentage of vanadium in magnetoflux and drehestahl.

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1. Introduction

Neutron Activation Analysis (NAA) is a nuclear analytical technique for non-destructive testing of materials, primarily used to determine a sample's elemental composition. It is done by irradiating the sample material with neutrons, causing it to become radioactive. Resulting in isotope decay and emitting gamma rays, which are then measured and analysed to quantify the elemental composition.

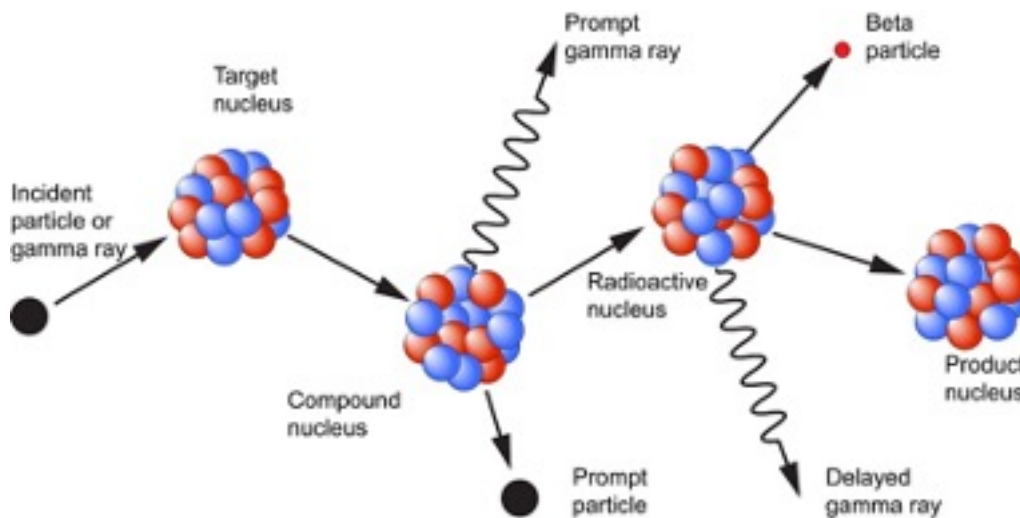


Figure 1: Diagram illustrating the process of neutron capture by a target nucleus followed by the emission of gamma rays (https://archaeometry.missouri.edu/naa_technical.html)

A neutron irradiates a Target nucleus. After capturing the incident neutron, the sample nucleus forms a compound nucleus, which then transforms into a radioactive, unstable nucleus and undergoes decay to reach stability. The radioactive nucleus emits a beta particle as a part of its decay process and some energy in gamma rays.

The samples used in the following experiment are **Vanadium (V)** and **Copper (Cu)**

Fission Sources: ^{252}Cf ^{247}Am - ^9Be

2. Physical Principle

2.1 Neutron Capture

Neutron Capture (NC) is the process where a free neutron is absorbed by an atomic nucleus, resulting in a heavier isotope or an excited state of the nucleus. NC can be divided into two types:

- **Radiative Capture:** In radiative neutron capture, the neutron is captured by the nucleus, and the excess energy from this process is released as gamma radiation.
- **Non-Radiative Capture:** In non-radiative neutron capture, the capture of a neutron by the nucleus leads to a reaction that does not primarily emit gamma radiation. Instead, the energy is released in other forms, such as the emission of particles or through other processes.

The use of neutrons in **Neutron Activation Analysis (NAA)** is favoured over other particles like electrons for several reasons. Unlike neutrons, electrons carry a charge and experience repulsive electric forces when interacting with the sample, preventing thorough irradiation. The electron's charge also means that more energy would be needed to irradiate the sample effectively. Neutrons, on the other hand, have a neutral charge, allowing for deep penetration into the sample, inducing neutron capture reactions with minimal background interference. This makes neutrons ideal for the non-destructive analysis of a wide range of elements and isotopes in NAA.

2.2 Radioactivity

Radioactivity is the spontaneous emission of radiation from the nuclei of certain unstable atoms as they undergo decay to become more stable. This process, known as radioactive decay, involves the spontaneous conversion of unstable isotopes into other isotopes, releasing energy in the form of particles or electromagnetic waves termed ionizing radiation. The emission of energy from an unstable atomic nucleus characterizes this reaction

- **Types of Decay**

- ❖ **Alpha Decay:** Alpha decay is a form of radioactive decay in which an unstable atomic nucleus releases an alpha particle, comprising two protons and two neutrons (essentially a helium-4 nucleus). This emission alters the parent nucleus, resulting in a new element with a mass number decreased by 4 and an atomic number decreased by 2.

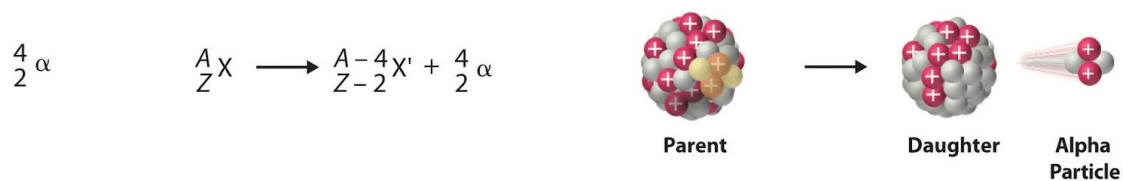


Figure 2: Alpha Decay visualization

- ❖ **Beta Decay:** Beta decay is a form of radioactive decay in which an unstable atomic nucleus releases a beta particle (either an energetic electron or positron), resulting in the transformation into an isobar of the original nuclide.



Figure 3: Beta Decay visualization

- ❖ **Gamma Decay:** Gamma decay is a form of radioactive decay in which an atomic nucleus moves from a high-energy, excited state to a lower-energy state by emitting a gamma ray photon.



Figure 4: Gamma Decay visualization

• Karlsruhe Nuclide Chart

The Karlsruhe Nuclide Chart

A nuclide chart is a two dimensional representation of the nuclear and radioactive properties of all known atoms. A nuclide is the generic name for atoms characterized by the constituent protons and neutrons. The nuclide chart arranges nuclides according to the number of protons (vertical axis) and neutrons (horizontal axis) in the nucleus. Each nuclide in the chart is represented by a box containing the element symbol and mass number, half-life, decay types and decay energies, etc.

"Magic" numbers

In nuclear physics, a magic number is a number of protons or neutrons (e.g. 2, 8, 20, 28, 50, 82, 126) which give rise to a complete shell in the atomic nucleus. Lead 208 for example, which consists of 82 protons and 126 neutrons, is called "doubly magic" since both the proton and neutron numbers are "magic".

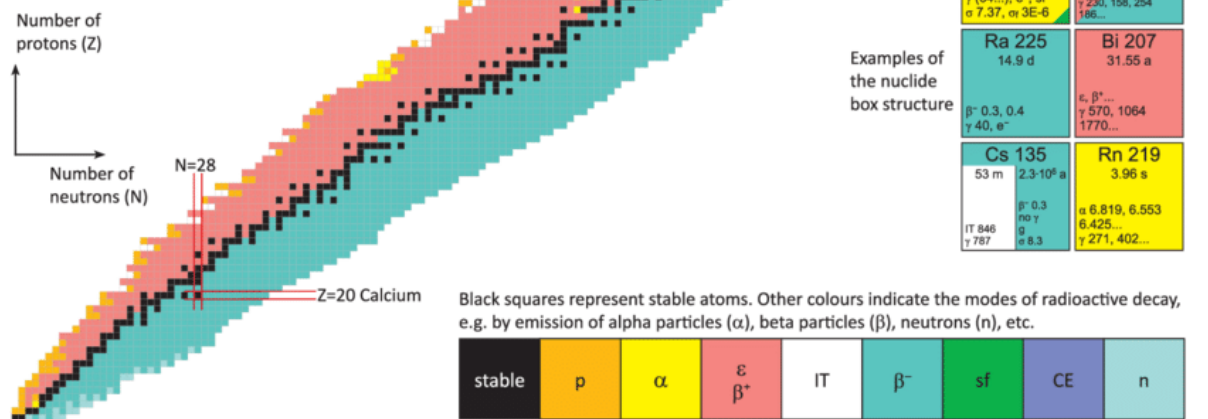


Figure 5: Karlsruhe Nuclide Chart (Zsolt Soti et.al (2018). Karlsruhe Nuclide Chart - New 10th Edition 2018)

The Karlsruhe Nuclide Chart is an extensive reference table that organizes nuclear data for experimentally observed nuclides (isotopes of elements). Key points include:

- ❖ Overview: This two-dimensional chart is arranged with neutron number (N) on the x-axis and proton number (Z) on the y-axis.
- ❖ Nuclide Representation: Each nuclide is depicted in a box that displays its element symbol, mass number (A), half-life, decay modes, energies of emitted radiation, and natural isotopic abundances.
- ❖ Color Coding: The boxes are colour-coded to indicate different types of radioactive decay, such as alpha, beta, gamma, and spontaneous fission.

2.3 Activation and Activity

2.3.1 Activation

Activation, also known as Neutron Activation, involves inducing nuclear excitation in a specific isotope through various stimulation channels or processes. The aim of activation is to trigger nuclear reactions within the isotope, resulting in outcomes such as energy release or the creation of new isotopes. Utilizing multiple stimulation channels enhances the likelihood of successfully exciting the target isotope.

2.3.2 Activation Energy and Activation Rate

Activation Energy, in the context of Neutron Activation, is the minimum energy a neutron needs to start or induce a nuclear reaction. This minimum energy typically represents the energy barrier that must be surpassed for the reaction to occur.

Activation rate refers to the frequency at which nuclear reactions occur via neutron activation, indicating the number of reactions per unit of time. The activation rate is influenced by several factors, including neutron flux, the cross-sectional area for the nuclear reaction, neutron energy, and the quantity and type of target atoms.

2.3.3 Activity

Activity refers to the rate at which radionuclides, produced through the activation process, decay. It is typically measured in becquerels (Bq). The activity, represented as $C(t)$, can be determined using the following equation

$$C(t) = \lambda B(t)$$

$B(t)$: Number of existing radionuclides

λ : Decay Constant

2.3.4 Formulas

During Activation, the following applies:

$$\begin{aligned} dB(t) / dt &= C_{\infty} - \lambda B(t) \\ \text{with Ansatz: } B(t) &= C_{\infty} / \lambda \times (1 - e^{-\lambda t}) \\ \text{it follows: } C(t) &= C_{\infty} \times (1 - e^{-\lambda t}) \\ \text{After the activation is stopped } t > t_B: C_{\infty} &= 0 \\ B(t) &= B(t_B) \times e^{-\lambda(t-t_B)} \end{aligned}$$

2.3.5 Activation: Technical Realisation

- The sample is bombarded with various types of radiation and particles to induce desired nuclear reactions
- Bombardment Of sample by
 - ❖ Gamma Radiation
 - ❖ High-energy electron beams
 - ❖ Charged particles like protons and deuterons
- Neutrons; preferably thermal neutrons with kinetic energy ranging from 0.01 to 0.1 eV for their favourable properties
- Nuclear Reaction of the Same Type (n, γ) Compound: The thermal neutrons are captured by the nucleus, leading to the emission of gamma rays.
- Large Cross Section → Good Sensitivity: The probability of neutron capture is high, which increases the sensitivity of the detection.
- Deep Penetration → Analysis of Thick Samples: Thermal neutrons can penetrate deeply into the material, allowing the analysis of thicker samples.
- No Coulomb Repulsion: Since neutrons are uncharged, they do not experience Coulomb repulsion, making it easier for them to interact with the nucleus.

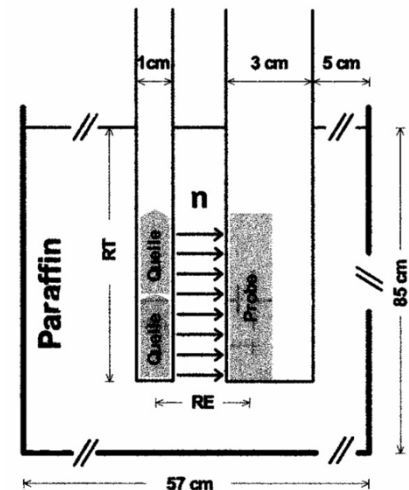


Figure 6: Activation of sample (Oleg Kalekin. Lab manual - Neutron Activation Analysis, FAU)

Sources used in the experiment are ^{252}Cf ^{247}Am - ^9Be as fission sources

Process summary:

- Neutrons are emitted from the fission sources
- The emitted neutrons are slowed down by a moderator to thermal energies
- Resulting nuclear reactions are analysed.

2.3.6 Activation of Vanadium

When a neutron is added to Vanadium (V), it converts into a proton, emitting an electron and an antineutrino (β^-). Both $^{50}\text{V}_{23}$ and $^{51}\text{V}_{23}$ are stable isotopes. Adding a neutron to $^{51}\text{V}_{23}$ produces $^{52}\text{V}_{23}$, which undergoes beta decay (β^-) to form $^{52}\text{Cr}_{24}$, a more stable isotope.

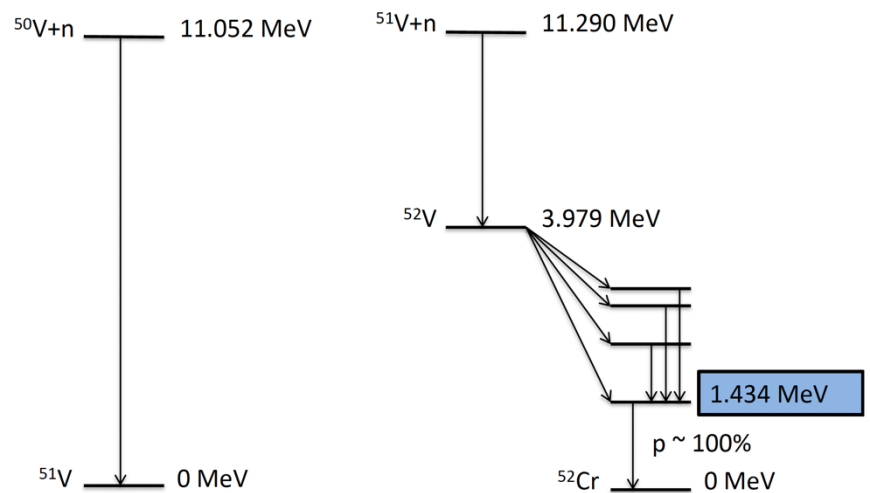


Figure 7: Activation of Vanadium
(Oleg Kalekin. Lab manual -
Neutron Activation Analysis, FAU)

2.3.7 Activation of Copper

Copper isotopes $^{63}\text{Cu}_{29}$ and $^{65}\text{Cu}_{29}$ are stable. However, when a neutron is added, they become radioactive, transforming into $^{64}\text{Cu}_{29}$ and $^{66}\text{Cu}_{29}$, respectively. $^{64}\text{Cu}_{29}$ further decays via both β^+ and β^- decay, producing $^{64}\text{Ni}_{28}$ and $^{64}\text{Zn}_{30}$. On the other hand, $^{66}\text{Cu}_{29}$, which has a longer half-life than $^{64}\text{Cu}_{29}$, undergoes β^- decay to form $^{66}\text{Zn}_{30}$.

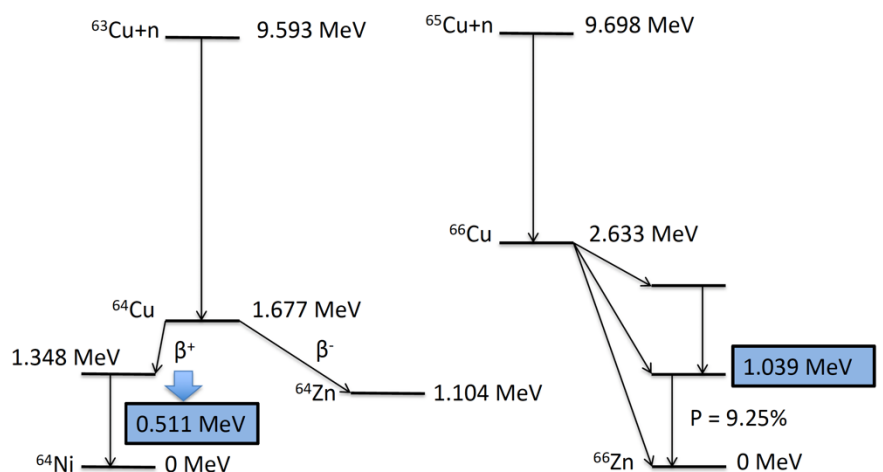


Figure 8: Activation of Copper (Oleg
Kalekin. Lab manual - Neutron
Activation Analysis, FAU)

2.4 Gamma Spectroscopy with Semiconductor Detectors

Gamma (γ) spectroscopy with semiconductor detectors, such as high-purity germanium (HPGe) detectors, is a powerful technique for measuring the energies and intensities of gamma rays emitted by radioactive sources.

2.5 Interaction of γ -radiation with matter

When gamma rays interact with matter, they primarily undergo three types of interactions:

2.5.1 Photoelectric Effect

- A gamma photon is completely absorbed by an atom, ejecting an electron from one of the inner shells (usually the K-shell).
- The energy of the ejected electron is equal to the energy of the incident gamma photon minus the binding energy of the electron.
- The probability of the photoelectric effect decreases with increasing gamma-ray energy and increases with atomic number Z .

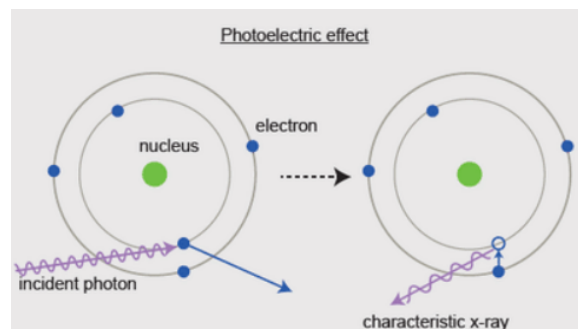


Figure 9: Photoelectric Effect (Lyu, Qihui & Neph, Ryan & Sheng, Ke. (2021). Pair production tomography imaging.)

2.5.2 Compton Scattering

- A gamma photon collides with a loosely bound outer-shell electron, transferring part of its energy to the electron and being scattered at a lower energy.
- The scattered photon retains some of the original energy, and the electron (called a Compton electron) is ejected from the atom.
- The probability of Compton scattering is inversely proportional to the gamma-ray energy and does not strongly depend on Z .

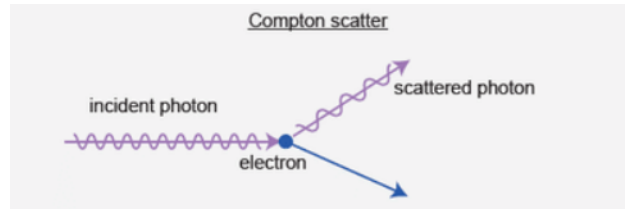


Figure 10: Compton Scattering (Lyu, Qihui & Neph, Ryan & Sheng, Ke. (2021). Pair production tomography imaging. 10.21203/rs.3.rs-151960/v1.)

2.5.3 Pair Production

- If the gamma-ray energy exceeds 1.022 MeV (the combined rest mass energy of an electron and a positron), the photon can be absorbed and converted into an electron-positron pair in the vicinity of the nucleus.
- The positron eventually annihilates with an electron, producing two 511 keV annihilation photons.
- The probability of pair production increases with gamma-ray energy and is proportional to Z^2 .

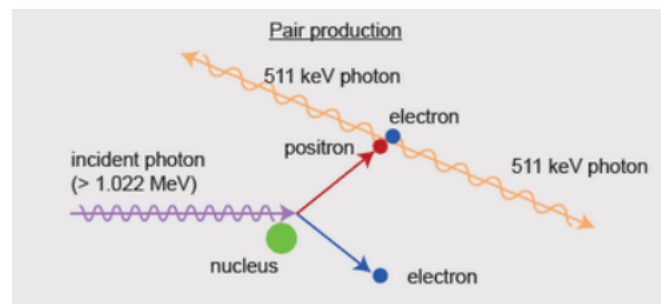


Figure 11: Pair Production (Lyu, Qihui & Neph, Ryan & Sheng, Ke. (2021). Pair production tomography imaging. 10.21203/rs.3.rs-151960/v1.)

2.5.4 Probability Curve/Graph

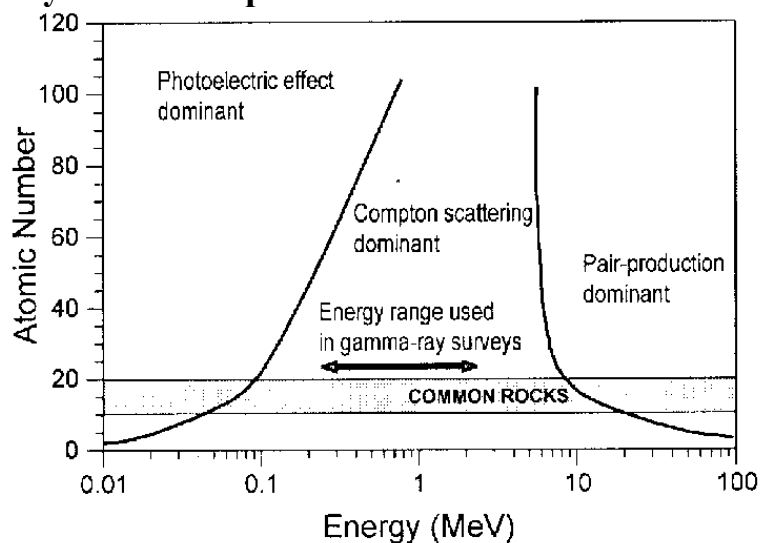


Figure 12: The interaction of gamma rays with matter. (Erdi-Krausz, G. & Matolin, Milan & Minty, Brian & Nicolet, J.P. & Schetselaar, Ernst. (2003). Guidelines for radioelement mapping using gamma ray spectrometry data.)

2.6 Absorption Law

The attenuation of gamma rays as they pass through a material is described by the exponential absorption law:

$$I(x) = I_0 e^{-\mu x}$$

where,

- $I(x)$ is the intensity of the gamma radiation at depth x .
- I_0 is the initial intensity of the gamma radiation.
- μ is the linear attenuation coefficient, which depends on the material and the energy of the gamma rays.
- x is the thickness of the absorbing material.

The attenuation coefficient μ is the sum of contributions from the photoelectric effect, Compton scattering, and pair production.

Element	Use	Isotope		σ_{act}/b	Radionuclide		
			A/%			$T_{1/2}$	E_γ/MeV
Vanadium	Tools	^{50}V	0.24	---	---	---	---
		^{51}V	99.76	4.5	^{52}V	3.75 min	1.434
Copper	Coins Brass	^{63}Cu	69.1	3.9	^{64}Cu	12.8 h	0.511
		^{65}Cu	30.9	1.8	^{66}Cu	5.1 min	1.039

Table 1: Activation data of test samples [1]

3 Detection

3.1 Experimental Setup

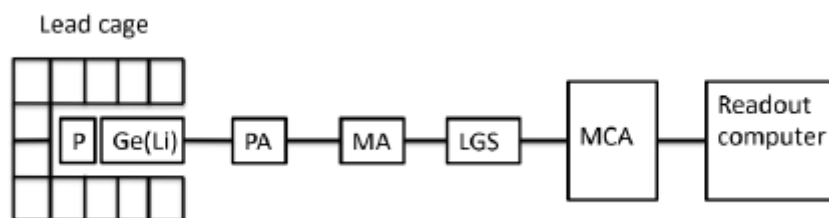


Figure 13: Experimental setup of the detector [1]

Here,

- P: Probe
- Ge(Li): Germanium (lithium-doped) semiconductor detector cooled with liquid nitrogen

- PA: Preamplifier
- MA: Main amplifier
- LGS: Linear gate stretcher
- MCA: Multi-channel analyser

3.2 Semiconductor Detectors

Semiconductor detectors, especially HPGe detectors, are favoured in gamma spectroscopy due to their superior energy resolution compared to scintillation detectors like NaI(Tl).

- **Detection Process:**

- ❖ When gamma rays interact with the semiconductor material, they generate electron-hole pairs proportional to the energy of the incident gamma photon.
- ❖ The applied electric field in the detector causes the electrons and holes to move towards the electrodes, creating a current pulse.

- **Energy Resolution:**

- ❖ The energy resolution of a detector is its ability to distinguish between gamma rays of different energies.
- ❖ HPGe detectors have high energy resolution, typically a few tenths of a percent at 1 MeV, due to the small amount of energy required to create an electron-hole pair (about 3 eV in germanium).

3.3 Detection Electronics

The signal generated in the semiconductor detector is processed by the following electronics:

- Pre-amplifier: Amplifies the initial charge pulse near the detector.
- Amplifier: Further amplifies and shapes the voltage pulse.
- Analog-to-Digital Converter (ADC): Converts the analog signal to a digital one corresponding to gamma-ray energy.
- Multichannel Analyzer (MCA): Sorts digital signals into an energy spectrum histogram.
- Data Acquisition System: Records and analyzes the energy spectrum to identify and quantify gamma rays.

3.4 Statistical Method

The energy spectrum is analysed to identify the radioactive isotopes in the sample and determine their concentrations. This is achieved by matching the peaks in the spectrum with known gamma-ray energies of specific isotopes. The intensity of these peaks indicates the concentration of each isotope.

To accomplish this, we extracted data from the MCA to a readout computer and plotted it using peak fitting software. We then fitted Gaussian curves to the peaks in our data.

$$G(x) = \frac{N_e}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} + C \quad (1)$$

- $G(x)$: Represents the Gaussian function, describing the data distribution along the x-axis.
- N (Count Rate): Determines the total count under the Gaussian curve.
- x : The independent variable at which the Gaussian distribution is evaluated.
- μ (Mean): The centre of the peak, indicating the peak's position along the x-axis.
- σ (Standard Deviation): Controls the width of the peak, measuring data deviation from the mean.
- C : The constant term, representing the baseline or noise in the data.

When we fit a Gaussian curve to the peaks, we adjust the parameters N , μ , and σ to find the best-fit Gaussian distribution that accurately represents the shape of the peaks in our data.

For calibrating error, we used Gaussian Error Propagation:

$$\Delta f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 \cdot (\Delta x)^2 + \left(\frac{\partial f}{\partial y}\right)^2 \cdot (\Delta y)^2 + \dots} \quad (2)$$

where Δf represents the uncertainty or error in the final quantity f

To determine the concentration of elements (Cu and V) in the sample, we use the following formula:

$$conc_{unknown} = \frac{\left(\frac{m_{std\ sample} \cdot Ne_{unknown}}{Ne_{std\ sample}} \right)}{m_{unknown}} \cdot 100\% \quad (3)$$

4. Analysis

4.1 ^{22}Na spectrum (Calibration spectrum)

When we obtain readings from the multichannel analyser in pulse height analysis (PHA) mode, we receive the counts per unit time associated with a specific channel number. We must identify the corresponding energies concerning the channel numbers to calibrate the energy. Energy calibration is a vital part of any experiment report, especially in the context of a Neutron Activation Analysis (NAA) experiment. In these experiments, energy calibration involves aligning the recorded gamma-ray energies with their known actual energies. Specifically, in this experiment, using a gamma calibration source like ^{22}Na , which emits gamma rays at 511 keV and 1275 keV, is essential for establishing a reliable energy scale. This calibration ensures that the Multi-Channel Analyzer operates accurately in Pulse Height Analysis (PHA) mode, correctly associating the data collected during the 10-minute acquisition time with the corresponding energy levels. We use the spectrum of ^{22}Na to identify the known energy peaks on the plot as shown in figure 14.

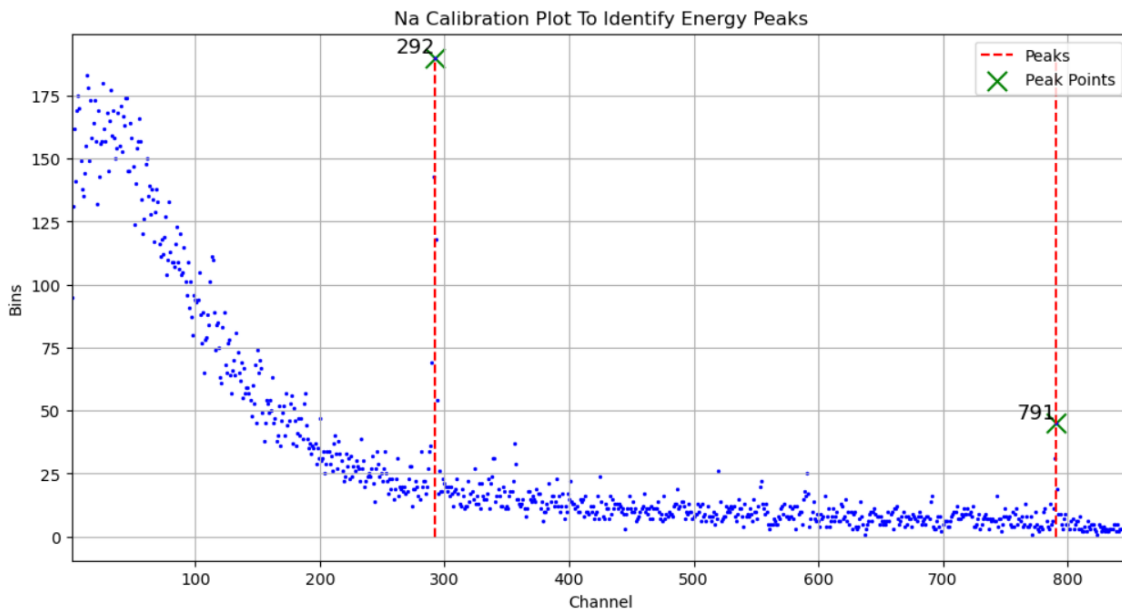


Figure 14: ^{22}Na energy peaks

From the plot above, we can clearly identify the distinctive ^{22}Na peaks at 511 keV and 1275 keV. The Gaussian fits of these peaks are shown below.

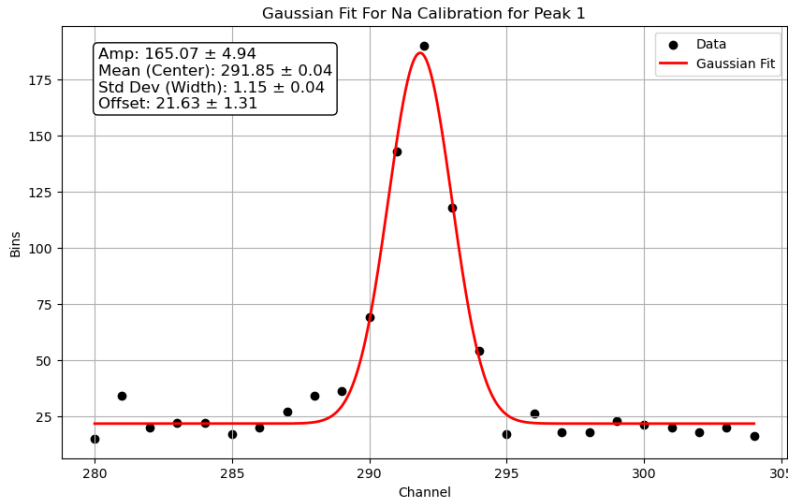


Figure 15: ^{22}Na peak 1 at channel 291 corresponding to 511 keV

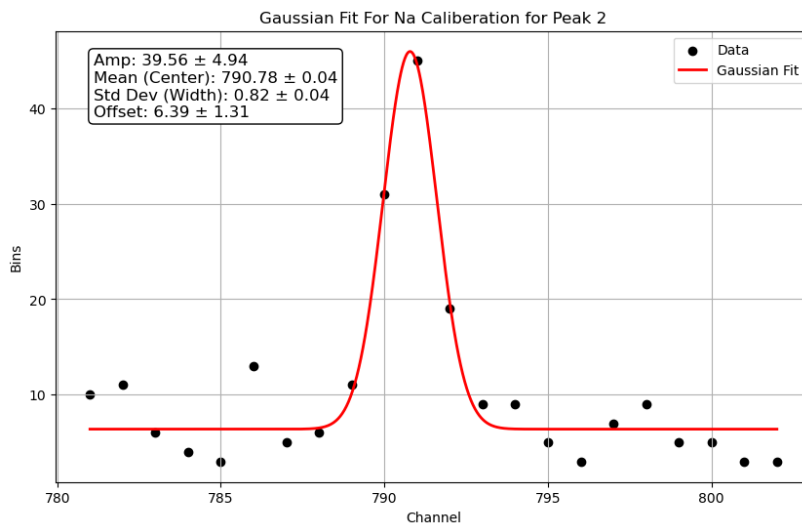


Figure 16: ^{22}Na peak 2 at channel 790 corresponding to 1275 keV

Energy (keV)	μ	σ	c	N_e
511	291.85 ± 0.04	1.15 ± 0.04	21.63 ± 1.31	186.69 ± 5.11
1275	790.78 ± 0.04	0.82 ± 0.04	6.39 ± 1.31	45.95 ± 3.09

Table 2: Fit parameters for ^{22}Na Gaussian curve.

4.2 Energy calibration using ^{22}Na Spectrum

Accurate energy calibration is crucial for identifying and quantifying elements in the sample, which is fundamental for the success and validity of the NAA experiment and, therefore, the reliability of the experiment report.

For this experiment, we describe the energy dependence of the channel number in the spectrum using a first-degree polynomial:

$$E_{\gamma} = A + B \cdot Ch = \underline{63.92} + \underline{1.53 \cdot Ch} \quad (4)$$

where,

error must be calculated, the rounding

- E_{γ} : Gamma-ray energy,
- Ch : Spectral channel number for the peak centre (usually the channel with the maximum counts),
- A : Intercept, the gamma-ray energy when Ch is zero,
- B : Slope, indicating how E_{γ} increases with each change in Ch , responsible for converting channel numbers into actual gamma-ray energies.

Performing linear regression yields a calibration curve for the first-order polynomial, providing a linear relationship between the channel numbers and the corresponding energies.

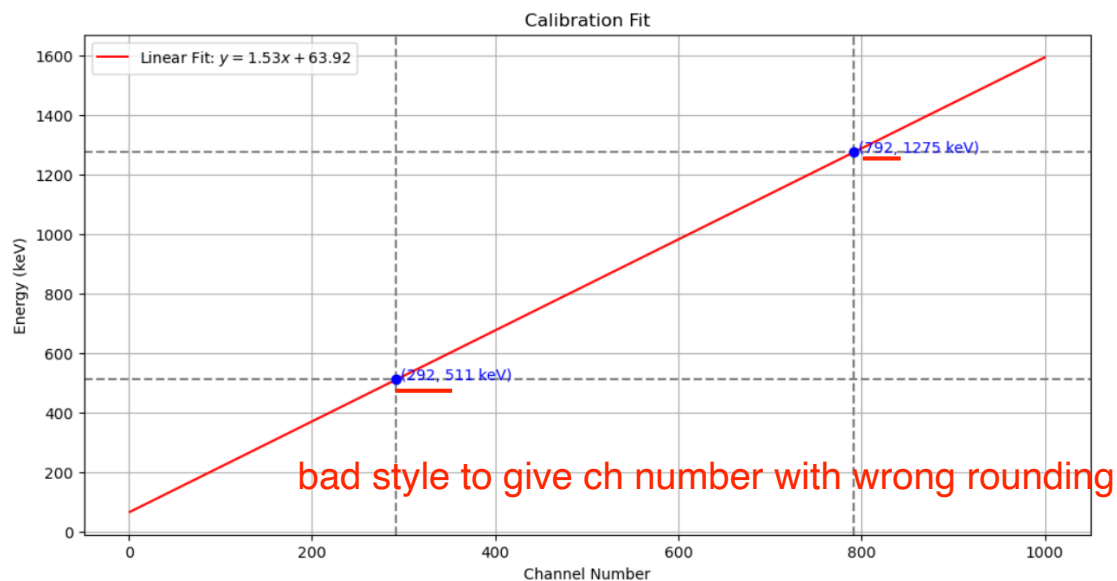


Figure 17: Energy calibration fit of Ge(Li) detector

rounding

Energy (keV)	FWHM (channels)	FWHM (keV)
511	2.7	4.1
1275	1.9	2.3

Table 3: Peak energies with FWHM values

The data obtained for ^{22}Na offer valuable insights into the energy resolution and Full Width at Half Maximum (FWHM) of the detected gamma-ray peaks. Post calibration, the FWHM for the peak at 511 keV was found to be 4.1463 keV, and for the peak at 1275 keV, it was 2.9529 keV. These FWHM values provide a measure of the detector's capability to differentiate between various energy levels in the gamma-ray spectrum. A lower FWHM indicates superior energy resolution, meaning the detector can more accurately distinguish between closely spaced energy peaks. The obtained FWHM values indicate a significant level of energy resolution, essential for precise gamma-ray spectroscopy and analysis.

rounding

4.3 Background Analysis

Measurement of the background was taken in two ways:

- With the lead cage door open at an angle of 90° (open box reading).
- With the lead cage door closed (closed box reading).

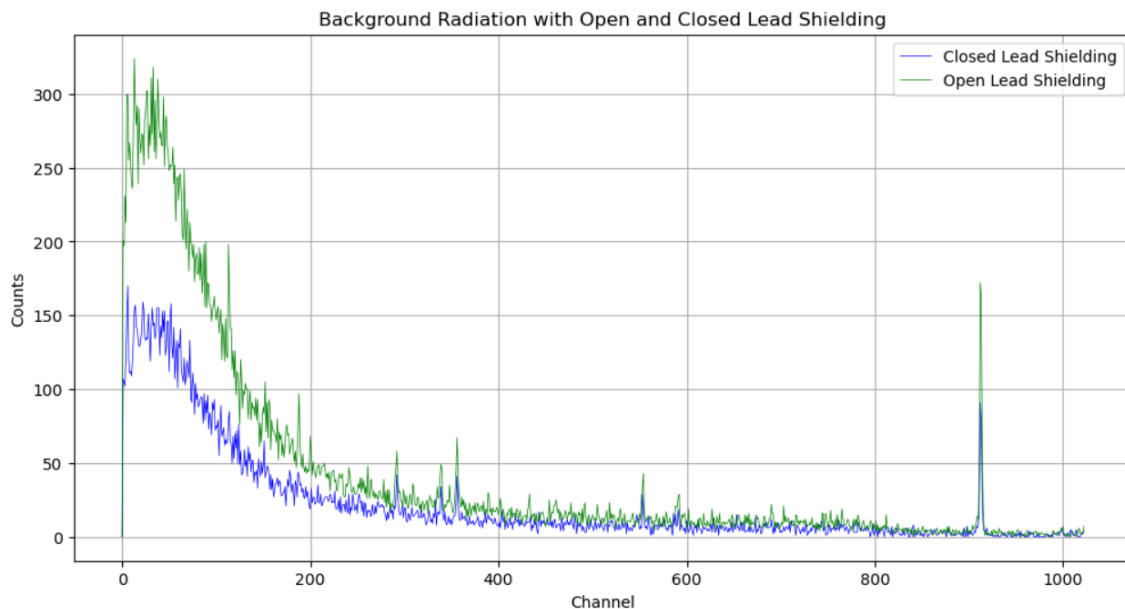


Figure 18: Comparison of background when the door is open and closed

Figure 18 shows the background differences in both scenarios, highlighting the lead shielding's effectiveness. The shielding, made from lead bricks, surrounded the Ge(Li) detector. This setup was crucial in reducing external radiation interference, thereby enhancing the spectral data's resolution and accuracy.

Effectiveness of lead shielding (%):

error, rounding

$$\frac{N_{e \text{ closed box}}}{N_{e \text{ open box}}} \cdot 100\% = \frac{23556}{44042} \cdot 100\% = 53.48\%$$

wrong definiton => wrong interpratation (46.5)

Therefore, we can conclude that the lead shielding reduces the background by around 53.48% when one side of the box is opened. Furthermore, the effectiveness of lead shielding can be assessed by considering the following factors:

- Shielding Solid Angle:

Calculate this solid angle (or just area)
Take proportion to the total one
Recalculate shielding efficiency

The shielded solid angle (SAA) represents the portion of the total solid angle in three-dimensional space from which gamma rays are directed towards the sample during Neutron Activation Analysis (NAA). In NAA, maximizing the solid angle from which gamma rays reach the sample is essential for improving analysis accuracy. Lead shielding helps by minimizing background radiation and the unintended activation of nearby materials, thereby reducing the SAA, and ensuring more neutrons are directed at the sample. This process reduces systematic errors in the experiment.

The effectiveness of the shielding and the overall geometry of the NAA setup affect the shielded solid angle. For our analysis, we utilized lead blocks with dimensions of 20 x 10 x 5 cm, resulting in an inner box area of 900 cm². A larger setup area allows for a wider solid angle, enhancing the accuracy and sensitivity of the analysis by increasing the number of interactions.

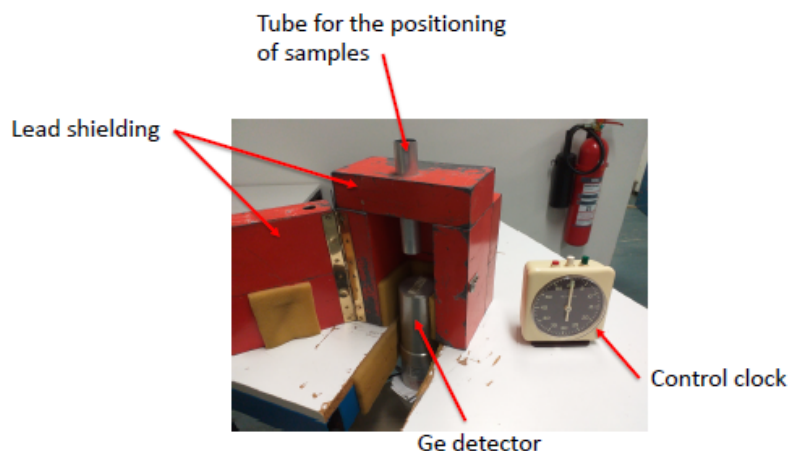


Figure 19: Lead box used in the experiment (Oleg Kalekin. Lab manual - Neutron Activation Analysis, FAU.)

- Energy Dependence of Absorption in Lead: **must be calculated**

The energy dependence of absorption in lead refers to how effectively lead shields against various energy levels. In the open-door configuration, the observed peaks displayed a significant increase in intensity and background noise, highlighting the impact of external radiation sources. These peaks were wider and less distinct, making precise measurement and analysis more difficult.

On the other hand, in the closed-door configuration, the peaks showed greatly improved resolution and lower background noise. This aligns with expectations, as the lead shielding efficiently blocked external radiation, reducing interference and improving the detector's accuracy.

Bgr peaks identification

4.4 Copper (Cu)

4.4.1 Standard Sample

Copper naturally occurs in two different isotopes: ^{63}Cu and ^{65}Cu . Consequently, 69.1% of copper is composed of the isotope ^{63}Cu , while 30.9% is composed of the isotope ^{65}Cu . To ascertain which peak corresponds to Cu in the gamma-ray spectrum of our standard Cu sample, we compared it to the peaks in the closed-box reading spectrum. We detected two distinct peaks for Cu at channel numbers 292 and 637, which were absent in the closed box reading spectrum. ~~Consequently, we concluded that these peaks represent Cu.~~

wrong

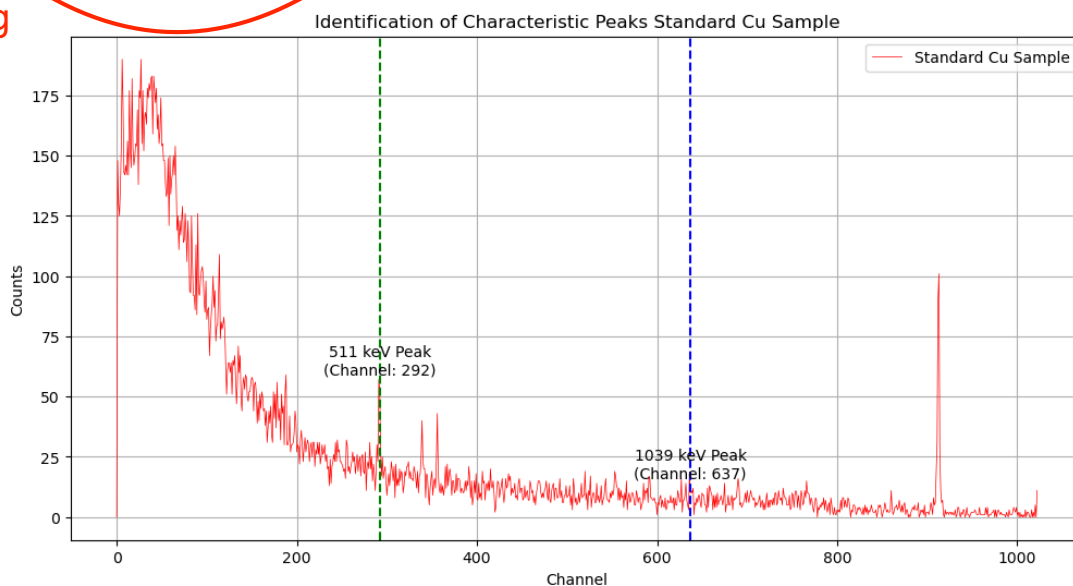


Figure 20: Identification of characteristic peaks for standard Cu sample

The peak at channel number 292 (511 keV) from the ^{63}Cu isotope cannot be used, as this transition has a half-life of over 12 hours and also it corresponds to both Na and Cu, so we only use the peak at channel number 637 (1039 keV).

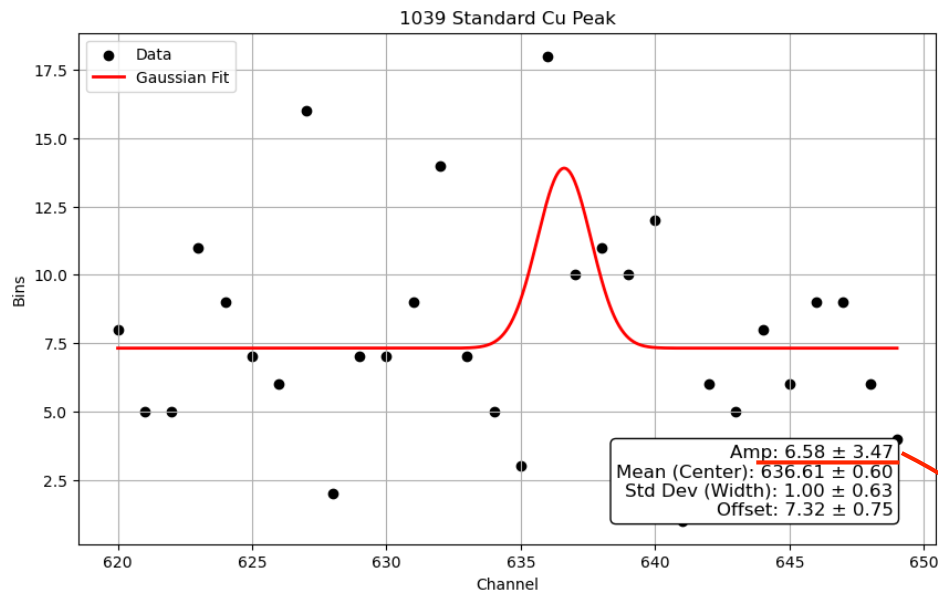


Figure 21: Gaussian fit of Cu peak at 1039 keV

Peak	μ	σ	c	N_e
1039 keV	<u>636.61 ± 0.60</u>	<u>1.00 ± 0.63</u>	<u>7.32 ± 0.75</u>	<u>13.90</u>

Table 4: Fit parameters of Cu standard sample

fit is bad, maybe not converge, can't be used in this form

We apply Poisson Statistics, which are based on the standard deviation of background measurements:

$$M_{min} = 2\sigma_B \quad (5)$$

$$\sigma_B = \sqrt{B}, \text{ where } B = kc \quad (6)$$

where:

- M_{min} : Minimum detected rate
- σ_B : The standard deviation of the background
- B : Background
- k: Number of channels = 20 why?
- c: Offset = 7.32 ± 0.75

Final result for min mass

Poisson statistics help us understand that the number of gamma rays detected in a given time interval fluctuates randomly. These fluctuations can result in statistical errors when determining the sample's elemental composition.

4.4.2 1€ Coins Sample

To pinpoint the peaks in the sample of 1-euro coins, we compared the gamma-ray spectrum peaks of the coins with those in the gamma-ray spectrum of the Standard Cu sample.

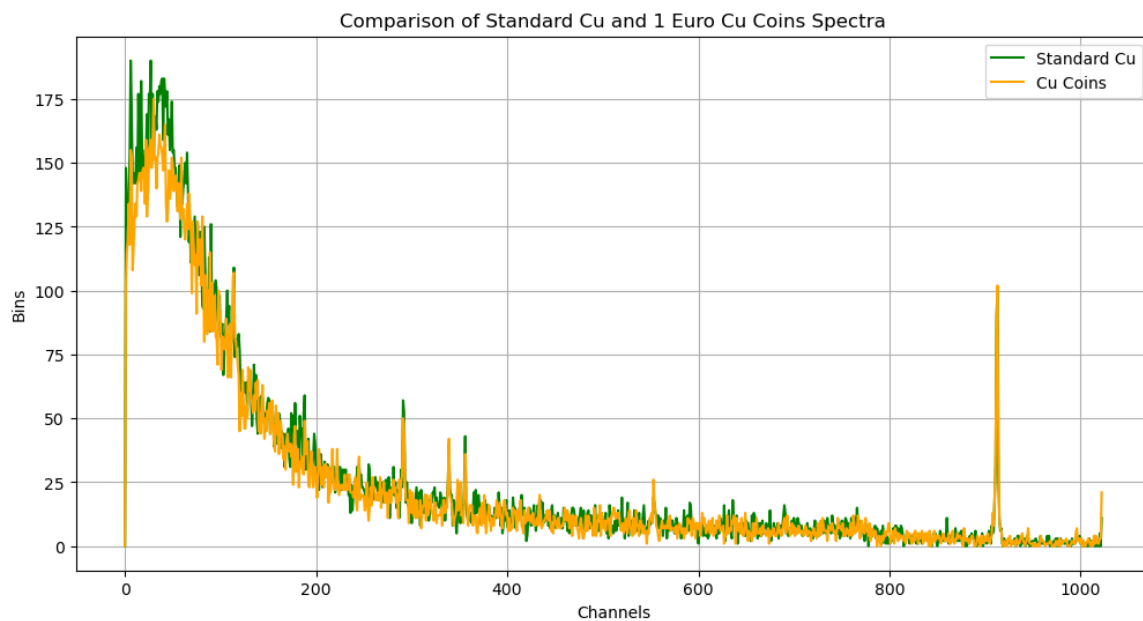


Figure 22: Plot comparison of standard Cu sample and Cu coins (1€) sample

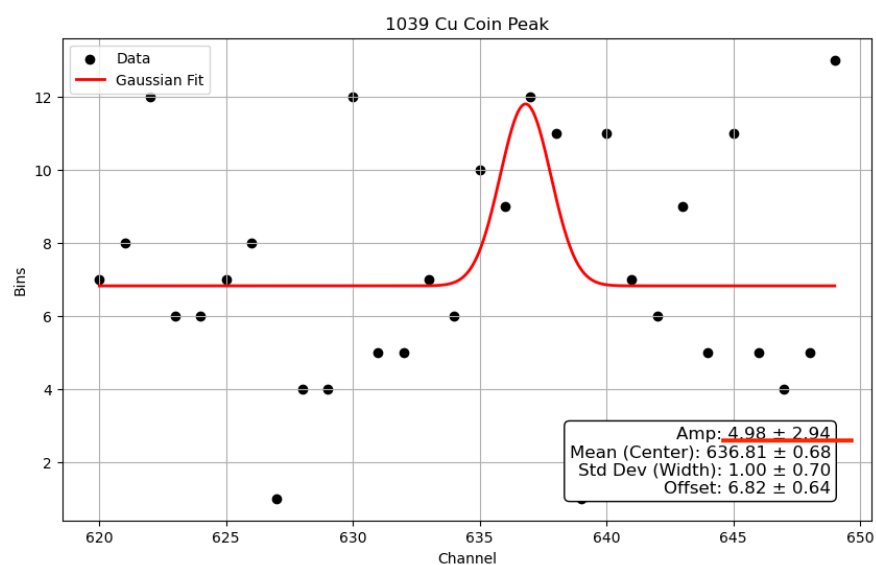


Figure 23: Gaussian fit of Cu peak at 1039 keV for 1€ coin

Peak	μ	σ	c	N_e
1039 keV	<u>636.81 ± 0.68</u>	<u>1.00 ± 0.70</u>	<u>6.82 ± 0.64</u>	<u>11.79</u>

Table 5: Fit parameters of 1€ sample

Using equation (3), we calculated the concentration of Cu in a given sample of 1€ coins to be $75.68\% \pm 60\%$ compared to the 75% suggested by the literature. The unexpected concentration of error value may result from errors in mass measurements, instrument calibration, sample preparation, environmental factors, or human error. Further investigation is needed to identify and address these issues.

4.5 Vanadium (V)

4.5.1 Standard Sample

Similar to Cu, to ascertain which peak corresponds to V in the gamma-ray spectrum of our standard V sample, we compared it to the peaks in the closed-box reading spectrum. The Vanadium content in our sample is 15.67%. Therefore, the amount of pure vanadium in the 16.53g sample is 2.6g.

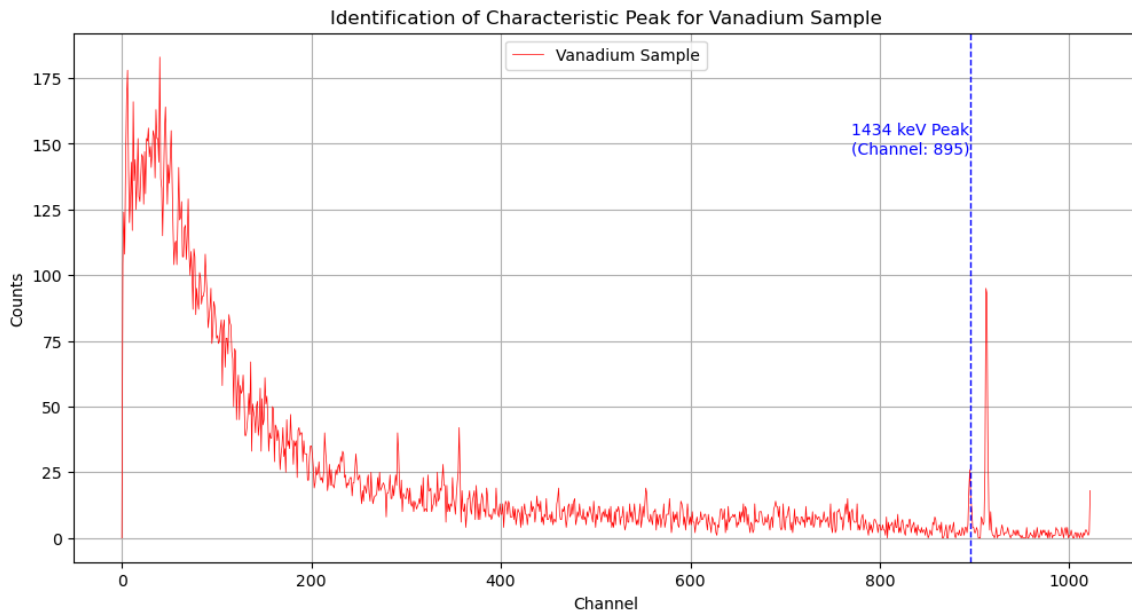


Figure 24: Identification of characteristic peaks for standard V sample

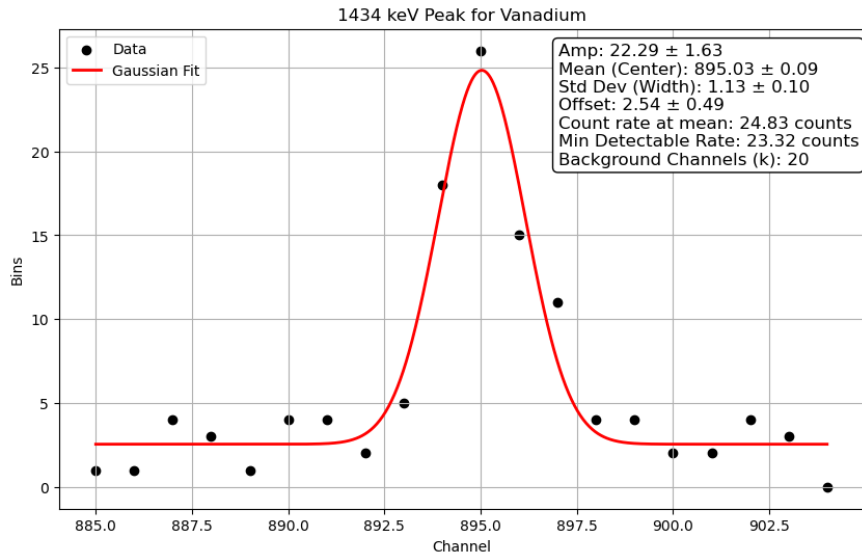


Figure 25: Gaussian fit of V peak at 1434 keV for the standard sample

Peak	μ	σ	c	N_e
1434 keV	895.03 ± 0.09	1.13 ± 0.10	<u>2.54 ± 0.49</u>	<u>24.83</u>

Table 6: Fit parameters of standard V sample

min mass

4.5.2 Two types of Steel containing Vanadium – Magnetoflex and Drehstahl

We identify the V peak in the Magnetoflex and Drehstahl samples by comparing it with the standard V sample.

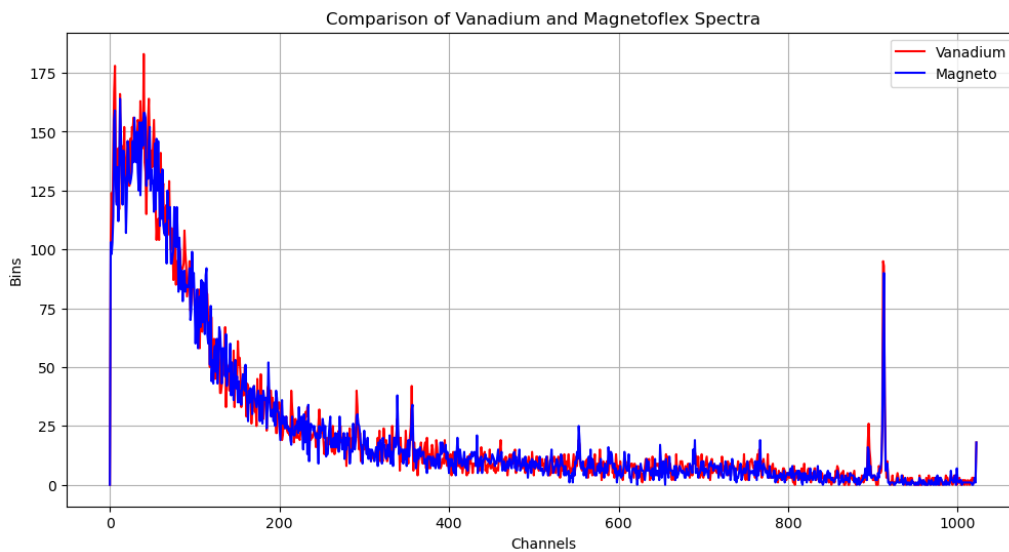


Figure 26: Plot comparison of standard Vanadium sample and Magnetoflex

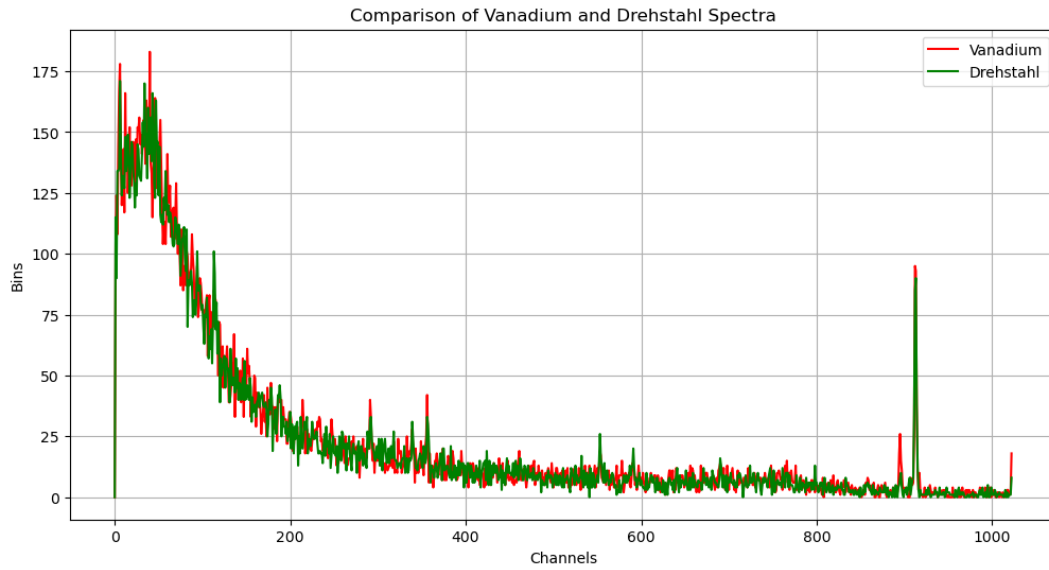


Figure 27: Plot comparison of standard Vanadium sample and Drehstahl

After which, the Gaussian fitting was obtained as below,

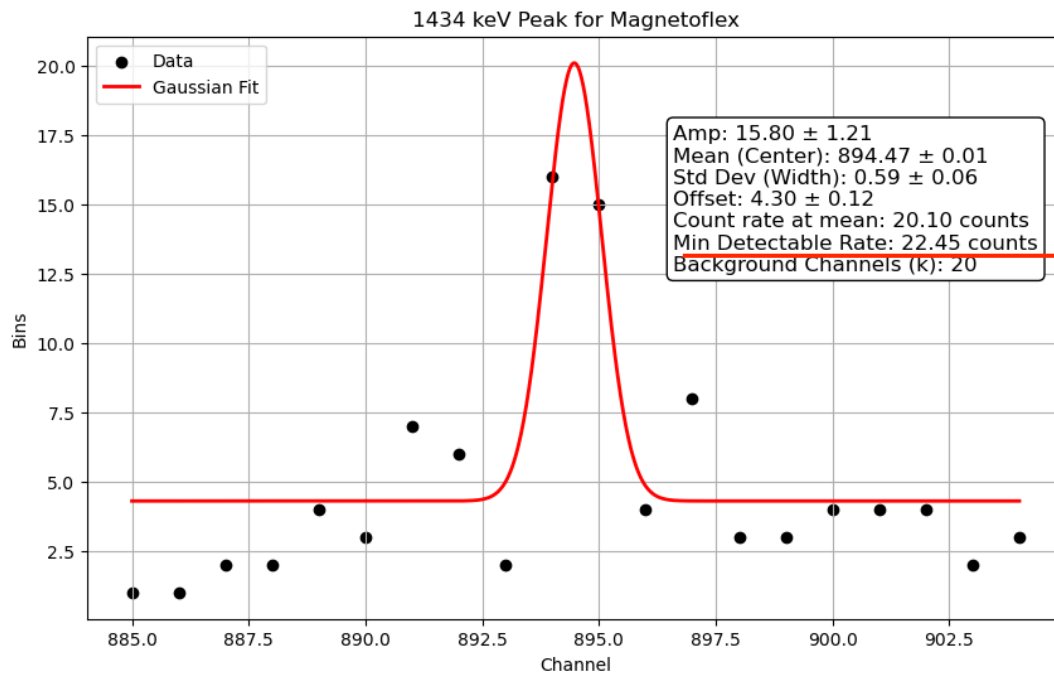


Figure 28: Gaussian fit of V peak at 1434 keV for Magnetoflex

Peak	μ	σ	c	N_e
1434 keV	<u>894.47 ± 0.01</u>	0.59 ± 0.06	4.30 ± 0.12	<u>20.10</u>

Table 7: Fit parameters of Magnetoflex

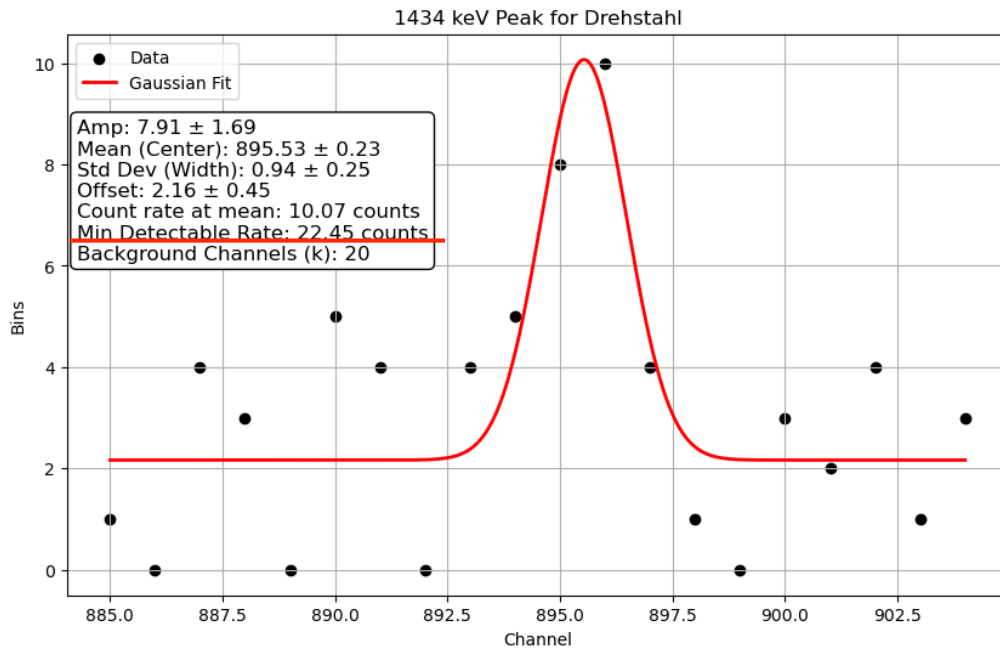


Figure 29: Gaussian fit of V peak at 1434 keV for Drehstahl

Peak	μ	σ	c	N_e
1434 keV	895.53 ± 0.23	0.94 ± 0.25	<u>2.16 ± 0.45</u>	<u>10.07</u>

Table 8: Fit parameters of Drehstahl

With the help of equation (3), we obtained the mass and % of Vanadium present as,

Steel	mass of V	% of V
Magnetoflex	0.77 ± 0.19 g	<u>$3.28\% \pm 0.82\%$</u>
Drehstahl	0.61 ± 0.25 g	<u>$2.31\% \pm 0.95\%$</u>

Table 9: Calculated mass of Vanadium in Steel samples

4.5.3 Half-life of Vanadium

The experiment included measuring the half-life of the isotope ^{52}V using a Multi-Channel Analyzer (MCA) set to Multi-Channel Scaling (MCS) mode. In this mode, the MCA logs all incoming events within a designated time interval (here, 10 seconds) into a single channel. Following this interval, it switches to a different channel and records all events in the subsequent time interval. We anticipate observing a decline in the number of counts over time because the quantity of activated Vanadium decreases.

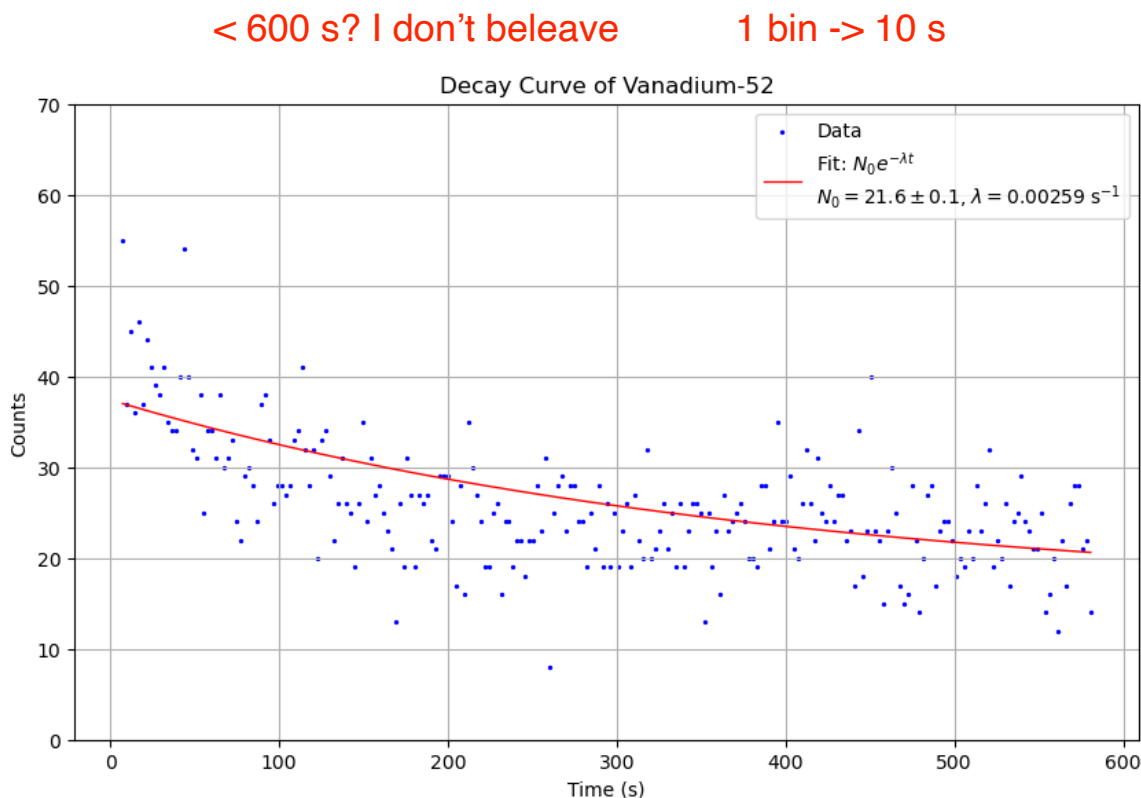


Figure 30: Decay curve of ^{52}V

The decay curve with the number of decays as a function of time is stated as,

$$N(t) = N_0 e^{-\lambda t} \quad (7)$$

where,

wrong, bgr is not counted

- N_0 : The initial number of radionuclides
- $N(t)$: Remaining number of radionuclides at time t
- λ : Decay constant

$$t_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$$

where, $t_{\frac{1}{2}}$ is half life

errors!

λ	0.00259 s^{-1}
$t_{\frac{1}{2}}$	<u>$268 \pm 0.69 \text{ s}$</u>

Table 10: Half-life of ^{52}V

Hence the recorded half-life of ^{52}V in our experiment is $268 \pm 0.69 \text{ s}$ or 4.47 minutes, whereas the literature value is 3.75 minutes.

5. References:

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